Research Project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Anglysis & Self Healing

Summary of RP work carried out during the reporting period.

Date	Work
28.03.2026	Started planning the basic workflow and component structure of the system.
01.04.2026	Collected and reviewed research papers related to failure root cause analysis and self healing techniques.
06.04.2026	Collected and explored CI/CD pipeline failure logs dataset
11.04.2026	Got team meeting with supervisor to discuss research progress

Supervisor's comments ÷ currently working on the component
Student processing well in research project : (Yes) / No
If no :
Project work to be carried out before next evolution period:
Improve the model accuracies.

Code based checked : Yes / No

Signature : [signature]
Date : 20/04/2026

CamScanner

Research project periodic log entry form

Project ID : : R26 - SE - 039
Student ID : IT22371768
Name : Ranasinghe S I
Individual Component : Failure Analysis & Self Healing.

Summary of RP work carried out during the reporting period.

Date	Work
19.04.2026	Refined research problem and finalized the scope of the Failure Analysis and Self Healing component.
20.04.2026	Started designing the high-level system architecture and identified main modules of proposed system.
22.04.2026	Researched suitable machine learning algorithms for root cause classification and compared Random Forest and Gradient Boosting techniques.
26.04.2026	Defined data collection strategy and identified required test execution artifacts such as logs, stack traces, screenshots.
29.04.2026	Developed the mock frontend for the system.

Supervisor's comments ÷ currently completing the 30% Scope.
Student processing well in research project : (Yes) / No
If No :
Project work to be carried out before next evolution period :

Code based checked : Yes / No

Signature :
Date : 30/04/2026

Date ___/___/___ Pg ___

Research project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self Healing

Summary of RP work carried out during the reporting period.

Date	Work
03.05.2026	Started preprocessing the collected CI/CD pipeline failure logs and stack traces for feature extraction.
06.05.2026	Applied TF-IDF vectorization to the textual log data.
08.05.2026	Developed the initial data labeling schema for categorizing root causes.
10.05.2026	Set up the python development environment using scikit-learn to begin training the Random Forest classification model.
11.05.2026	Conducted a team review to verify the accuracy of the manually labled failure instances for the training dataset

Supervisor's comments : works well in the research.
Student processing well in research project : (Yes) / No
If NO :
Project work to be carried out before next evolution period :

Code based checked : (Yes) / NO

Signature : Sa
Date : 17/06/2026

Date / /

Research project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self Healing

Summary of RP work corrected out during the reporting period.

Date	Work
19.05.2026	Finalized the feature engineering process by integrating structured features such as exception types and retry counts
21.05.2026	Completed the comparative analysis between Random Forest and Gradient Boosting models for root cause classification
25.05.2026	Evaluated the classification model using performance metrics including accuracy, precision and F1-Score.
27.05.2026	Developed the initial logic for Decision Analysis Module to map identified root causes to specific repair actions
30.05.2026	Started designing the DOM analyzes component

Supervisor's Comments: Progress well in the research ..
Student processing well in research project : (Yes) / No
If no :
Project work to be carried out before next evolution period :

Code based checked : (Yes) / No

Signature : [redacted]
Date : 17/06/2026

RATNA

Research Project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S I
Individual Component : Failure Analysis & Self Healing

Summary of RP work carried out during the reporting period.

Date	work
02.06.2026	Studied different self-healing approaches used in software systems.
05.06.2026	Researched how MCP can connect LLMs with Github repositories
08.06.2026	Investigated methods for automatically analyzing source code using LLMs.
10.06.2026	Designed the initial workflow for the self-healing component
15.06.2026	Conducted a progress review meeting with supervisor.

Supervisor's Comments : works well in the research.
Student processing well in research project : (Yes) / No
If no :
Project work to be carried out before next evalution period :

Code based checked : (Yes) / No

Signature :
Date : 17/06/2026

Date ___/___/___ No: ____

Research Project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self Healing

Summary of RP work carried out during the reporting period.

Date	work
16.06.2026	Designed the detailed self-healing workflow
18.06.2026	Studied how LLMs can identify software defects from source code.
20.06.2026	Planned the automated code correction process
23.06.2026	Defined the validation mechanism after applying fixes

Supervisor's comments : works well in the research.
Student processing well in research project : (Yes) / No
If No :
Project work to be carried out before next evolution period :

Code based checked : (Yes) / No

Signature : ____
Date : 26/06/2026

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Date ___/___/___ Pg ___

Research Project Periodic log & entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self Healing.

Summary of RP work carried out during the reporting period.

Date	Work
03.07.2026	Finalized the root cause categories and reviewed the preprocessing logic for CI/CD failure data.
07.07.2026	Integrated the trained root cause classification model with the Failure Analysis backend service.
11.07.2026	Developed the initial decision logic to map identified root causes into suitable recovery actions
15.07.2026	Implemented database models to store failure records, healing actions, notifications and analysis results
18.07.2026	Tested the failure analysis workflow using sample error messages, stack traces and pipeline logs.

Supervisor's comments :
Student processing well in research project : Yes / No
If No :
Project work to be carried out before next evolution period :

Code based checked : (Yes) / No

Signature :
Date : 27/07/2026

RATHNA

Research Project periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self Healing.

Summary of RP work carried out during the reporting period.

Date	Work
21.07.2026	Defined repair eligibility rules for application defects using confidence, repository and source file information
24.07.2026	Developed read-only repair planning workflow using repair agent, LLM and GitHub MCP integration.
27.07.2026	Implemented safety checks for repository, branch, commit SHA, file path and protected source files.
29.07.2026	Developed the developer-approved repair publishing flow using using a separate auto-heal branch and draft pull request.
31.07.2026	Tested the end to end failure analysis and controlled self healing workflow and fixed backend integration issues

Supervisor's Comments :
Student processing well in research project : Yes / No
If No :
Project work to be carried out before next evolution period:

Code based checked : (Yes) / No

Signature:
Date : 25/08/2026

Date / / No

Research Project Periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self Healing

Summary of RP work carried out during the reporting period.

Date	Work
04.08.2026	Started integrating Github Actions with the Failure Analysis component to discover failed workflow run automatically.
08.08.2026	Implemented project-scoped retrieval of failed Github workflow runs and failed jobs using the selected project's configuration
12.08.2026	Developed secure Github job-log retrieval with log sanitization, secret redaction and failure evidence extracti
16.08.2026	Integrated Github failure evidence with the trained 9 class root cause classification model.

Supervisor's Comments :
Student processing well in research project : Yes / No
If No :
Project work to be carried Out before next evolution period :

Code based checked : (Yes) / No

Signature :
Date : 29/08/2026

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Research Project Periodic log entry form

Project ID : R26-SE-039
Student ID : I122371768
Name : Ranasinghe S.I
Individual Component: Failure Analysis & Self Healing.

Summary of RP work carried out during the reporting period.

Date	Work
22.08.2026	Tested the Github based failure analysis flow using a real Python syntax failure and verified application defect
24.08.2026	Improved the frontend to display failed CI runs, analysis results, controlled repair actions and
26.08.2026	Extended failure evidence extraction to support Java / Maven compilation errors, long workflow logs.
27.08.2026	Tested Java compilation failures and refined the confidence-based recovery policy for controlled repair
28.08.2026	Improved project-scoped Github credential handling for decision the repair agent and investigated repair planning and publishing validation issues.

Supervisor's Comments:
Student processing well in research project : Yes / No
If No :
Project work to be carried out before next evolution period:

Code based checked : (Yes) / No

Signature :
Date : 29/08/2026

Research Project Periodic log entry

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self Healing.

Summary of RP work carried out during the reporting period

Date	Work
03.09.2026	Improved Github Actions failure log processing and extraction of relevant diagnostic information
07.09.2026	Refined the 9-class root cause prediction workflow and reviewed classification results
11.09.2026	Improved failure analysis backend integration with recovery recommendations and notifications
15.09.2026	Reviewed the controlled repair workflow and developer approval requirements.
18.09.2026	Tested failure analysis using sample CI/CD logs and investigated identified backend issues.

Supervisor's Comments :
Student proresssng well in research project : (Yes) / No
If No :
Project work to be carried out before next evaluation period :

Code based checked : (Yes) / No

Signature : [signature]
Date : 19/10/2026

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Date ___/___/___ No. ___

Research Project Periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371762
Name : Ranasinghe S.J
Individual Component : Failure Analysis & self healing

Summary of RP work carried out during the reporting period

Date	Work
21.09.2026	Refined eligibility checks for application defects before allowing controlled repair planning
23.09.2026	Reviewed repository safety checks, protected file restrictions and separate approval steps.
28.09.2026	Tested selected failure analysis and repair agent functions and reviewed the results.
30.09.2026	Investigated remaining integration limitations and documented ~~requirment~~ required improvements.

Supervisor's Comments.
Student processing well in research project : (Yes) / No
If No :
Project work to be carried out before next evaluation period :

Code based checked : (Yes) / No

Signature : S
Date : 19/10/2026

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Date ___/___/___

Research Project Periodic log entry form

Project ID : B26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self healing

Summary of RP work carried out during the reporting period.

Date	Work
02.10.2026	Reviewed the final machine learning model and documented nine-class classification results.
05.10.2026	Analyzed model evaluation metrics including accuracy, precision, recall and F1-score.
08.10.2026	Reviewed the complete failure analysis and controlled self healing system architecture.
11.10.2026	Updated research documentation covering implementation, evaluation, system limitations

Supervisor's Comments :
Student processing well in research project : (Yes) / No
If No :
Project work to be carried out before next evaluation period :

Code based checked : (Yes) / No

Signature :
Date : 19/10/2026

Date ___/___/___ NF ___

Research Project Periodic log entry form

Project ID : R26-SE-039
Student ID : IT22371768
Name : Ranasinghe S.I
Individual Component : Failure Analysis & Self healing.

Summary of RP work carried out during the reporting period

Date	work
15.10.2026	Reviewed final thesis content and refined the explanation of research contributions
16.10.2026	Preparing Component 3 demostration covering failure classification and recovery actions.
17.10.2026	Reviewed repair agent safety controls, developer approval and draft pull request workflow.
19.10.2026	Prepared industry expert validation questions and reviewed the system demostration.
20.10.2026	Prepared for the final research viva and revised implementation, results, limitations

Supervisor's Comments :
Student processing well in Research project : (Yes) / No
If No :
Project work to be carried out before next evaluation Period :

Code based checked : (Yes) / No Signature :
Date : 20/10/2026